

Experiment log — 2026-09-04 to 2026-09-18

Every change tried against the volume error, what it was measured against, and what it did. Nothing here is committed to the pipeline; every row is an offline experiment on the run outputs, so the pipeline's own numbers are unchanged throughout. Ground truth is tape unless a row says water.

The one number to keep in mind: on a rigid slim can the pipeline's girth is ****+0.9%**** against calipers. Everything above that on a limb is limb, scale, or wrap – not the model's ability to measure a circumference.

1. Marker detection

change	tried on	result	kept?
White and black cords instead of green	`sunshine\_v2`	Stage 0 (colour-agnostic detector) boxed both bands on 7/7 frames; Stage 3's colour rule could not separate a neutral cord from skin; every plane rejected, no cut	no – cords re-dyed green
Per-band colour instead of one capture average	`sunshine\_v2`	Averaging white + black gave grey RGB[71,64,64]; per-band gave RGB[151,140,142] and [41,34,32]	reverted – moot once cords were green
Keep the projected plane when its colour twin is later gated out	`sunshine\_v2`, cohort 0–6	Recovered the cut on `sunshine\_v2`; on the cohort the same defect had destroyed the lower band on `0\_left`, `2\_left`, `5\_right` (sliver / runaway volumes)	reverted at the user's request; defect still present
Drop foot slices from the limb-axis fit	`sunshine\_v2`	Real ankle band went from 35.7° (rejected at 35°) to 26.1°	reverted with the above

2. Ground truth

change	result
Water displacement (weigh tub before/after)	Carries two biases: film on the withdrawn limb (+20–50 g, 7–17% of an ankle reading) and fill-level shortfall below the overflow point (–100 cm³ per mm on a leg-sized tub). Used on captures 0–6 and `test5`
Circumference tape, disc model, 2 cm spacing	Chosen as primary. Two passes on `test5`: 0.8 cm mean / 2 cm worst on girth, 2 cm on band separation (7% of volume). Water and tape agreed to 4% on `test5` (2070 vs 1985)
Two-point cone through the band girths only	Under-reads the segment by 12–22% (misses the calf bulge). Not usable as truth
Circle-from-circumference bias	Over-reads area by 1.3% at a/b 1.2, 2.6% at 1.3, 6.3% at 1.5. Limbs measured a/b 1.1–1.5. Handbook corrected from “~2%, constant”

3. Where the error is — per capture

Volumes are the span between the two bands unless noted. "wrap" is the alpha fallback's inflation, measured directly on `test6`.

capture	subject	girth err (low / high band)	volume err	alpha fallback	notes
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`test1`	M, hairy	+3.1% / +10.5%	no volume truth	no	four capture styles gave the same error
`test2`	M	+6.8% / +14.3%	no volume truth	yes, $\chi=-2$	worst residual, 4.08 mm
`test3`	M	+7.3% / +10.1%	no volume truth	no	cube $10.8 \times 10.9 \times 10.7$ – best cube
`test4`	M	+5.3% / +10.9%	no volume truth	no	100% ring coverage, still +10.9%
`test5`	M, hairy shin	+1.5% / +6.7%	**+24.9%** (water 2070)	yes, $\chi=-4$	profile: shin +19%, calf belly +4%
`test6`	nearly hairless	+7.9% / +13.2%	**+38.7%** (tape 1399)	yes, $\chi=-22$	profile flat +8–12%, no shin spike
`fanta_orange`	rigid can	**+0.9%** (calipers 18.5)	+5–10% (height +6.8%)	yes, $\chi=-12$	girth exact; height is end caps
`fanta_red`	can, mirror floor	–	15,164 cm ³	yes	floor removal took 19.5%; cube fill 0.359

`test6` decomposed exactly:

cube-derived scale	+4.2% linear	$\times 1.133$	ruler 27.5 vs pipeline 28.7 between bands
residual girth	+5.5%	$\times 1.114$	profile vs tape, scale removed
alpha wrap		$\times 1.10$	mesh section / points, summed over slices = 1.120
product		$\times 1.387$	measured +38.7%

4. Hypotheses tested and ruled out

hypothesis	test	verdict
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Upper cut plane sits too high	green band located from mesh colour vs plane height	plane within 0.05–0.7 cm of the band on every sound capture. **Wrong** – my own earlier claim
Girth over-read comes from incomplete ring coverage	coverage vs error across captures	`test4` at 100% coverage still +10.9%. **Wrong** – my own earlier claim
Leg shake during capture	error vs height (–0.32) and vs surface scatter (–0.19); four capture styles agree	anti-correlated with both. **Ruled out**
Stray outer points inflate the fitted ring	six trim rules at the band slices	shell is 15 mm thick with an *inward* tail; best rule moved girth 0.4%. **Ruled out**
Reconstruction method inflates girth	Poisson vs alpha_shape on `test1`	circumference byte-identical (it is read from the Stage 3 cloud); alpha volume +17% worse. **Ruled out for girth**
Vertical blur of the profile	Gaussian blur + gain fit to the tape profile	rms 1.78 → 1.52 cm; systematic +2 cm left on the shin. **Weak, not the cause**
Cube is physically $\neq$ 10 cm	can girth +0.9% on a scene whose cube read 0.815 fill	would have read +4% if the cube were 9.6. **Probably per-scene reconstruction, not the object** – caliper still pending
Hair	`test5` (hairy) vs `test6` (nearly hairless) profiles	shin spike +19% present only on `test5`. **Supported**

5. The wrap, seen directly

`ring\_gallery.png`: `test6` sliced every 2 cm with the watertight mesh's cross-section drawn over the cleaned points.

h cm girth mesh area / points' ellipse
0-12 20-27 1.03 - 1.08
14-28 29-32 1.13 - 1.17 one side carries a second arc ~1 cm inside the
 outer one; the 30x alpha bridges the outer strays
summed over the span: 1.120 (+12.0% volume from the wrap alone)

Poisson tunnelled (χ from -2 to -22) on 5 of 8 runs, which is what forces the alpha fallback. A finer alpha search between the ladder's rungs found nothing tighter than the 25x it chose on `test5`.

6. MLS radius — the one change that moved the volume

Replays Stage 3's ghost chain from the pre-MLS cluster with only the MLS radius varied; then re-wraps. The can is the control.

CAN (girth 18.5, cylinder ≤ 394.9)				TEST6 (tape 1398.6)			
mls	girth	poisson χ	volume	girth	poisson χ	volume	
0	+0.7%	-6	—	+10.5%	-30	—	(alpha 2141)
4	+0.5%	-12	—	+10.0%	-6	—	(alpha 1982) <- pipeline today
8	+0.6%	-2	—	+9.3%	0	—	(alpha 1780)
12	+0.6%	2	384.2	+9.2%	2	1606.6	+14.9%
16	+0.6%	2	385.5	+9.3%	2	1615.7	+15.5%
24	+1.5%	2	392.0	+0.6%	-2	—	<- artefact, see below
32	+3.7%	2	407.9	-12.7%	-44	—	

****12-16x spacing closes Poisson on both objects**** — no fallback, no wrap — and takes `test6` from +38.7% to +15% with the can within 3% of its bound.
****Girth does not move**** at any radius up to 16x, which is what a merged double sheet should do: the merged ring lands between the two sheets, where the ellipse fit already was. The +15% that remains is cube scale and the real over-read, now visible instead of buried under the wrap.

The 24x row on `test6` reads +0.6% and looks like a fix. It is not: the can at the same radius moves the *other* way (+1.5%, then +3.7% at 32x) while `test6` collapses to -12.7%. Opposite signs on the two objects is distortion crossing zero by accident.

****Stacked passes**** (the pipeline's 4x first, then a wide pass):

CAN				TEST6			
mls	girth	poisson χ	volume	girth	poisson χ	volume	
4-12	+0.6%	-2	—	+9.7%	-4	—	
4-16	+0.6%	2	384.4	+9.6%	2	1614.5	+15.4%
4-24	+0.6%	2	386.4	+10.0%	2	1632.1	+16.7%

Same volume as the single 12-16x pass, same untouched girth – and more robust: `4→24` holds the can at +0.6% and `test6` at +10.0% where the single 24x pass distorted both. Denoising each sheet first evidently keeps the wide pass from tearing the surface. ****4→16` is the candidate\*\***: Poisson closes on both objects, the can lands at 384 against a 395 bound, `test6` goes from +38.7% to +15.4%, and nothing about the girth is hidden.

What this change does and does not do, stated plainly: it removes the alpha fallback and its wrap inflation (the $\times 1.10$ term). It does not touch the cube scale ($\times 1.133$) or the residual girth ($\times 1.114$), which together are the +15% that remains. Those need the caliper and a different fix respectively.

****Cold run through the real pipeline**** (wired in behind `MLS\_SECOND\_RADIUS\_MULT`, default off, run with it set to 16; full detail in `COLD\_RUN\_REPORT.md`):

		today (4x)		cold run (4→16)		truth
test6	volume	1939.4	+38.7%	1703.9	+21.8%	1398.6 tape
	recon	alpha 30x		Poisson $\chi=2$		
	mesh/pts	1.105		0.989		
	girth	20.14	/ 32.72	19.95	/ 32.09	19.0 / 28.0
can	volume	415.5		387.2		≤ 394.9
	height	15.48	+6.8%	14.78	+1.9%	14.5
	girth	18.45		18.45		18.5

Poisson closed in the pipeline's own path on both objects; the cube was byte-identical between runs; the offline sweep's per-slice sections matched the real run to 0.001 but its volume was 5% low. Girth moved about 1% inward.

7. Environment failures found along the way

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| failure | symptom | cause | fix |
|---|---|---|---|
| Reflective floor | `fanta_red` 15 L "can", cube fill 0.359 | RANSAC removed 19.5% of the floor instead of ~70%; reflections form a second sheet | shoot on the matte tile |
| WSL GPU degraded | inference 14 s → 320–520 s, GPU 100% / 7.9 GB with nothing else on it | not memory, not swap (both checked and wrong at first) | `wsl --shutdown`; it also recovered on its own overnight |
| Background job from another repo | ~2.7 GB RAM held by a headless Blender render | `SCG/auto_entity_extraction` runner, state `T` (suspended) | killed; unrelated to this project |
| Stage 0 refuses a frame | `test5` img7 "cube missing" | cube occluded behind the ankle in that view | correct rejection; ran on the 7 clean frames |
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8. Still open

- ****Caliper the cube edge.**** Decides whether ``REFERENCE_REAL_SIZE_CM`` should change. Worth 13% of every volume.
- ****Re-shoot `fanta\_red` on the matte tile**** with a ruler on the band separation – the span-on-a-cylinder test.
- ****The projected-plane merge defect**** (section 1) is still in the pipeline and destroyed three of fourteen cohort captures.
- ****`1\_left` drifted 1.2% between identical runs**** on 2026-09-04; ``test6`` reproduced to the byte on 2026-09-18. Not understood.